AP Precalculus: 2026–2027

Unit 4: Functions Involving Parameters, Vectors, and Matrices   Lesson 4.2: Parametric Functions Modeling Planar Motion   (2 instructional periods, 55 minutes each)

Primary Objective: Students will identify key characteristics of parametric planar motion — horizontal and vertical extrema from the maximum and minimum values of $x(t)$ and $y(t)$, and axis intercepts of the curve from zeros of $x(t)$ and $y(t)$ — and apply these to describe a particle's position and range of motion. *(Big Idea: Functions)*

Priority Ideas & Skills

AP Skills	Key Understandings (EKs)
3.A *Describe characteristics* — Identify horizontal and vertical extrema of a parametric curve from the maximum and minimum values of the component functions $x(t)$ and $y(t)$; describe what those extrema tell us about the range of motion in each direction.	**EK 4.2.A.1**: A parametric function $f(t) = (x(t), y(t))$ models particle motion in the plane. The graph indicates the position of the particle at time t.
3.B *Apply numerical results* — Use the max/min values of $x(t)$ and $y(t)$ (found analytically or from a table) to bound the particle's motion, and use zeros of $x(t)$ and $y(t)$ to locate where the curve crosses the axes.	**EK 4.2.A.2**: The horizontal and vertical extrema of the particle's motion are the maximum and minimum values of $x(t)$ and $y(t)$, respectively.
	EK 4.2.A.3: Real zeros of $x(t)$ correspond to y-intercepts of the curve; real zeros of $y(t)$ correspond to x-intercepts.

Vocabulary, Concepts, & Theorems

Term	Definition / Note
horizontal extremum	The maximum or minimum value of $x(t)$ over the domain; determines the rightmost and leftmost positions of the particle.
vertical extremum	The maximum or minimum value of $y(t)$ over the domain; determines the highest and lowest positions of the particle.
y-intercept (of parametric curve)	A point on the curve where $x(t) = 0$; found by solving $x(t) = 0$ for t, then evaluating $y(t)$ at those values.
x-intercept (of parametric curve)	A point on the curve where $y(t) = 0$; found by solving $y(t) = 0$ for t, then evaluating $x(t)$ at those values.

Activate Prior Knowledge & Spiral Review

Spiral topics activated during Warm-Up (first 8 minutes):

- **Evaluating component functions**: Substitute t-values into $x(t)$ and $y(t)$ — the core computation from Lesson 4.1, now used to build intuition about extrema from a table.
- **Maximum and minimum of a quadratic**: Use the vertex formula $t = -b/(2a)$ to locate the max/min of $at^2 + bt + c$; this is the primary analytical tool for finding extrema.
- **Solving quadratic equations**: Set $x(t) = 0$ or $y(t) = 0$ and solve by factoring or the quadratic formula to find axis intercepts of the parametric curve.

Hook — Entry Event

Scenario: A ball is launched and its position (in feet) at time t seconds is modeled by:

$$x(t) = -16t^2 + 48t, \qquad y(t) = 12t - t^2.$$

Discussion prompts:

1. *"How high does the ball get? Which component function do you look at to answer that?"*
2. *"How far left or right does the ball travel? Which component tells you that?"*
3. *"When does the ball hit the ground (return to $y = 0$)? What equation do you solve?"*

Key tension to surface: Each question maps to one of three tasks — finding max/min of $x(t)$, finding max/min of $y(t)$, or solving $y(t) = 0$. These are exactly the skills of LO 4.2.A.

Transition: "Today we develop a systematic way to find horizontal extrema from $x(t)$, vertical extrema from $y(t)$, and axis intercepts from the zeros of each component."

Lesson — Instruction Sequence

Step 1 — Review & Planar Motion (5 min)
Briefly recall that $f(t) = (x(t), y(t))$ gives a particle's position at time t. Now ask: *how far* does it travel in each direction? Write EK 4.2.A.1 on the board. Use the hook scenario to frame the three questions the lesson answers.

Step 2 — Horizontal Extrema from $x(t)$ (12 min)
The leftmost and rightmost positions are the min and max of $x(t)$. Model with the hook: $x(t) = -16t^2+48t$ is a downward parabola. Vertex at $t = -48/(2\cdot(-16)) = 1.5$ s; $x(1.5) = 36$ ft (rightmost). EK 4.2.A.2. At $t = 0$, $x = 0$ (leftmost on $[0, 3]$).

Step 3 — Vertical Extrema from $y(t)$ (10 min)
Same technique for $y(t)$. Model: $y(t) = 12t-t^2$; vertex at $t = 6$, $y(6) = 36$ ft (highest). Identify $y(0) = 0$ as the starting height. Students practice on the second example in guided notes. EK 4.2.A.2.

Step 4 — Axis Intercepts from Zeros (15 min)
EK 4.2.A.3: zeros of $x(t)$ give y-intercepts; zeros of $y(t)$ give x-intercepts. Model: Solve $y(t) = t(12 - t) = 0 \Rightarrow t = 0$ or $t = 12$; find coordinates of each point. Solve $x(t) = -16t(t - 3) = 0 \Rightarrow t = 0$ or $t = 3$; students find coordinates. Period 1: Steps 1–2. Period 2: Steps 3–4 + activity.

Pacing note: Steps 1–2 in Period 1; Steps 3–4 + group activity in Period 2.

Active Monitoring

- **Mixing up components**: Students may find the max of $y(t)$ when asked for horizontal extrema. Prompt: *"Horizontal means left-right. Which axis is that? Which component controls it?"*
- **Reporting t instead of position**: Students stop at the t-value. Prompt: *"$t = 2$ is a time, not a position. What are the coordinates at that moment?"*
- **Sign errors in vertex formula**: Require students to write a, b, c explicitly before applying $t = -b/(2a)$.
- **Zeros vs. intercepts**: Students confuse zeros of $x(t)$ with x-intercepts. Post the rule: *zero of $x(t)$ → y-intercept; zero of $y(t)$ → x-intercept.*

Group Work & Differentiation

Students work in groups of 3–4 on the tiered activity (remote-control car scenario).

- **Tier R (Remediate)**: Given $f(t) = (2t,\ -t^2 + 4)$ for $0 \le t \le 3$, evaluate both components at $t = 0, 1, 2, 3$ and identify the leftmost/rightmost x-values and the highest/lowest y-values from the table. No algebra required.
- **Tier A (Accelerate)**: Given $f(t) = (t^2 - 4t,\ 3\sin(\pi t/2))$ for $0 \le t \le 4$, find horizontal extrema of $x(t)$ analytically (vertex) and vertical extrema of $y(t)$ by reasoning about the sine function's range.
- **Tier E (Extend)**: Compare two parametrizations of the same circular path — $f(t) = (\cos t, \sin t)$ and $g(t) = (\cos(2t), \sin(2t))$ — and determine whether they produce the same extrema, the same intercepts, and the same range of motion. Explain what changes and what stays the same.

Individual Work & Assessment

Exit ticket administered independently (final 5 minutes of Period 2).

1. Given $f(t) = (t^2 - 1,\ 2t - t^2)$ for $0 \le t \le 3$, complete a table at $t = 0, 1, 2, 3$.
2. Find the horizontal extrema. Show work using the vertex formula for $x(t) = t^2 - 1$.
3. For what values of t does the particle cross the y-axis? State the coordinates.

Data use: Students who cannot apply the vertex formula to $x(t)$ or $y(t)$ need re-teaching on quadratic max/min. Students who identify zeros of $y(t)$ as y-intercepts have the rule reversed — correct individually.

Reinforcement & Extension

Homework (Lesson 4.2): 5–6 problems covering finding horizontal and vertical extrema from component functions, locating axis intercepts of parametric curves, one contextual motion problem, and one AP-style multiple-choice item.

Preview of Lesson 4.3: Ask students: *"We described position with $(x(t), y(t))$. What if we also wanted to describe the* direction *of motion at any moment?"* Lesson 4.3 introduces vectors and connects parametric motion to vector notation.